QuantTerminal

Complete Project Documentation

A simple guide to the quantitative research terminal for Indian (NSE / BSE) and US stocks. Learn what it is, who built it, what every module does and how to use it, step by step.

Live app	https://quantterminal.streamlit.app/
Source code	github.com/MichaelF102/Quant_Terminal
Built with	Python and Streamlit
Markets	NSE, BSE, NYSE and NASDAQ

For educational purposes only. This dashboard is for educational purposes only. We are not liable for any investment decisions made by users.

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Disclaimer and How to Read This Document

Educational use only. QuantTerminal is built for learning and research. It does not give investment advice and it does not tell you what to buy or sell. This dashboard is for educational purposes only, and we are not liable for any investment decisions made by users.

Why the labels say Bullish, Bearish and Neutral

Many tools show Buy, Sell or Hold. QuantTerminal does not. It uses three neutral words that only *describe what the numbers show*:

Label	What it means	What it does NOT mean
Strongly Bullish / Bullish	Indicators lean upward right now	A promise the price will rise
Neutral	Indicators are mixed or flat	A promise nothing will change
Bearish / Strongly Bearish	Indicators lean downward right now	A promise the price will fall

How to read this document

- **Plain language:** each idea is explained in simple words first, then the technical name is shown.
- **Screenshots** were taken from the live app at quantterminal.streamlit.app.
- **Currency:** Indian stocks show the rupee sign, US stocks show the dollar sign.
- **Market data** means price and volume data collected from data providers such as Yahoo Finance.

Important limits to remember

- Past results do not guarantee future results.
- Every model is a simplification. It can be wrong.
- Data can be late, missing or revised by the provider.
- Always do your own research and talk to a qualified adviser before investing.

1 About This Project

What is QuantTerminal?

QuantTerminal is a free web app that lets you study the stock market using maths, statistics and machine learning. You pick a stock, and the app draws charts, runs tests, builds forecasts, simulates the future and can even write a 20-page research report for you.

The problem it solves

Professional analysis usually needs many separate tools: one for charts, one for statistics, one for backtesting, one for reports. They are expensive and hard to learn. QuantTerminal puts all of them in **one place**, with the same stock selected across every page.

Who is it for?

User	How they can use it
Students	Learn finance and data science with real market data
Beginner analysts	Practise research steps from charts to risk to reports
Researchers	Test ideas quickly and export results
Teachers	Show live examples in class

Core promises

- **No look-ahead bias:** tests only use information that was available at that time.
- **No fake numbers:** if data is missing, the app says so instead of inventing values.
- **Open method:** formulas and assumptions are written down in the docs folder.
- **Educational focus:** results describe history and do not give advice.

At a glance	
Modules	20 workspaces (19 analysis pages plus Credits)
Domains	7 research domains
Code size	About 67,000 lines of Python
Tests	23 automated test files
Data	Daily prices for NSE, BSE, NYSE and NASDAQ stocks

2 Key Features and What Makes It Special

The 7 research domains

Domain	What it covers	Modules
Market & Data	Prices, screening, indicators	Dashboard, Market Explorer, Technical Analysis
Statistics & Returns	Return behaviour and volatility	Statistical Analysis, Return Analytics, Volatility Lab
Forecasting	Predicting with statistics, ML and DL	Time Series, ML, DL Forecasting
Simulation & Risk	Possible futures and possible losses	Monte Carlo, Risk Analytics, Regime Detection
Strategy & Trading	Designing and testing rules	Strategy Lab, Backtesting, RL Trading
Portfolio & Research	Allocation and factor analysis	Portfolio Lab, Factor Research, Statistical Arbitrage
Reporting	Turning results into documents	Report Generation, Credits

The five featured workspaces (our special strengths)

Workspace	Why it stands out
Regime Detection	Finds calm and stormy market phases automatically using a Hidden Markov Model
Strategy Lab	Runs 11 strategies in a tournament and ranks them on a leaderboard
ML Forecasting	Compares tree ensembles and regressors with proper out-of-sample testing
DL Forecasting	LSTM and GRU neural networks for time-ordered data
RL Trading	A learning agent trained in a simulated market with an order log

Other highlights

- One sidebar controls the stock, region, exchange and time window for every page.
- Works for Indian and US markets with correct currency symbols.
- Exports to PDF, Excel and CSV.
- Runs in a normal web browser. Nothing to install for users of the live app.

3 Terminal Use Cases

Six common ways to use QuantTerminal.

Use case 1: Equity Research and Screening

Goal: Find and understand companies.

What you do

- Use Market Explorer presets to shortlist stocks.
- Open the Dashboard for price, valuation and returns.
- Check Technical Analysis for trend and momentum.

Example

A student wants large IT companies with rising momentum. They filter by sector and RSI, then compare two of them side by side.

Use case 2: Risk Measurement and Stress Testing

Goal: Understand how much could be lost.

What you do

- Read VaR and CVaR in Risk Analytics.
- Study drawdown depth and recovery time.
- Run Monte Carlo to see a range of outcomes.

Example

An analyst asks 'what is a bad month for this stock?' and reads the 95% VaR and the worst historical drawdown.

Use case 3: Strategy Design and Backtesting

Goal: Test a rule before believing in it.

What you do

- Build or pick a rule in Strategy Lab.
- Replay it in Backtesting with costs.
- Compare it with buy-and-hold.

Example

A learner tests an RSI rule and finds it loses after brokerage costs, which teaches why costs matter.

Terminal Use Cases

Use case 4: Portfolio Construction

Goal: Split money sensibly.

What you do
- Draw the efficient frontier in Portfolio Lab.
- Try Risk Parity weights.
- Study factor exposure and beta.

Example
A group project builds a 5-stock portfolio and compares equal weights with Max Sharpe weights.

Use case 5: Forecasting and Machine Learning

Goal: Compare prediction methods fairly.

What you do
- Fit ARIMA in Time Series.
- Compare ML and DL models out of sample.
- Train an RL agent and read its stance.

Example
A researcher sees that fancy models barely beat a simple baseline, an honest and useful lesson.

Use case 6: Learning and Academic Projects

Goal: Learn by doing.

What you do
- Study statistics with real data.
- See regimes and distributions visually.
- Export a 20-page report for coursework.

Example
A teacher shows fat tails in class using the Statistical Analysis page and a live stock.

In every use case the output describes **historical and statistical behaviour**. It is a learning aid and not a recommendation.

4 Roles, Credits and Governance

QuantTerminal was designed and developed by **Fincell Quants**. The work was split into three equal workstreams so that every member owned a clear part of the code.

Person	Background	Main role	Pages owned
Michael Fernandez	MSc Big Data Analytics	Lead engineer and architect: engine, strategies, backtesting, forecasting, ML, regimes, simulation	Backtesting, Forecasting, Machine Learning, Regime Detection, Monte Carlo, Settings
Pranav Sahai	2nd-year BSc Statistics	Statistics, volatility, risk, factors, stat-arb and exports	Return Analytics, Statistical Analysis, Volatility Lab, Risk Analytics, Factor Research, Statistical Arbitrage
Aashima Grover	2nd-year BSc IT	Technical indicators, charts, portfolio and reporting	Dashboard, Market Explorer, Technical Analysis, Portfolio Lab, Strategy Lab, Reports

Project leads	Pramit Datta and Vanshika Soni
Workstream balance	Michael 24 files, Pranav 23 files, Aashima 24 files (71 module files in total)
Team	Fincell Quants

How the project was planned (30 days)

Phase	Days	Focus
Phase 0	Half a day	Setup: virtual environment, branches, agree on function contracts
Phase 1-2	Days 1-14	Core maths, statistics, indicators, strategies and engines
Phase 3	Days 15-24	Build the Streamlit pages and connect them to the engines
Phase 4	Days 25-30	20-page PDF report, polish and cloud deployment

Rules the team followed

- Function signatures were frozen early in docs/ARCHITECTURE.md and docs/DATA_LAYER.md.
- Each person worked on their own branch and merged into a protected main branch.
- Each owner listed the 'bias controls' (such as leakage-proof ML) they had to respect.
- The licence is MIT and the code is hosted on GitHub.

5 System Requirements and Installation

Option A: use the live app (easiest)

Open **https://quantterminal.streamlit.app/** in any modern browser. Nothing to install. If the app has been idle it may need up to a minute to wake up.

Option B: run it on your own computer

Requirement	Details
Python	3.10, 3.11 or 3.12
Memory	8 GB recommended (4 GB works for light use)
Internet	Needed to download market data
Disk	About 3 GB because of TensorFlow and ML libraries

Step by step

- **1.** Download the code: `git clone https://github.com/MichaelF102/Quant_Terminal`
- **2.** Make a virtual environment: `python -m venv venv`
- **3.** Activate it: `source venv/bin/activate` (Windows: `venv\Scripts\activate`)
- **4.** Install the libraries: `pip install -r requirements.txt`
- **5.** Start the app: `streamlit run app.py`
- **6.** Your browser opens at `http://localhost:8501`

Main libraries

Purpose	Libraries
Web app	Streamlit
Data	yfinance, pandas, numpy, tradingview-screener
Charts	Plotly, Matplotlib, Seaborn
Statistics	statsmodels, scipy, arch
Machine learning	scikit-learn, XGBoost, LightGBM, CatBoost
Deep and reinforcement learning	TensorFlow, Gymnasium, TensorTrade
Optimisation	cvxpy, PyPortfolioOpt
PDF and Excel	ReportLab, openpyxl, pypdf

> **Tip:** if installation fails, upgrade pip first (`pip install --upgrade pip`) and use Python 3.12. NumPy must stay below version 2.0.

6 A Tour of the Screen

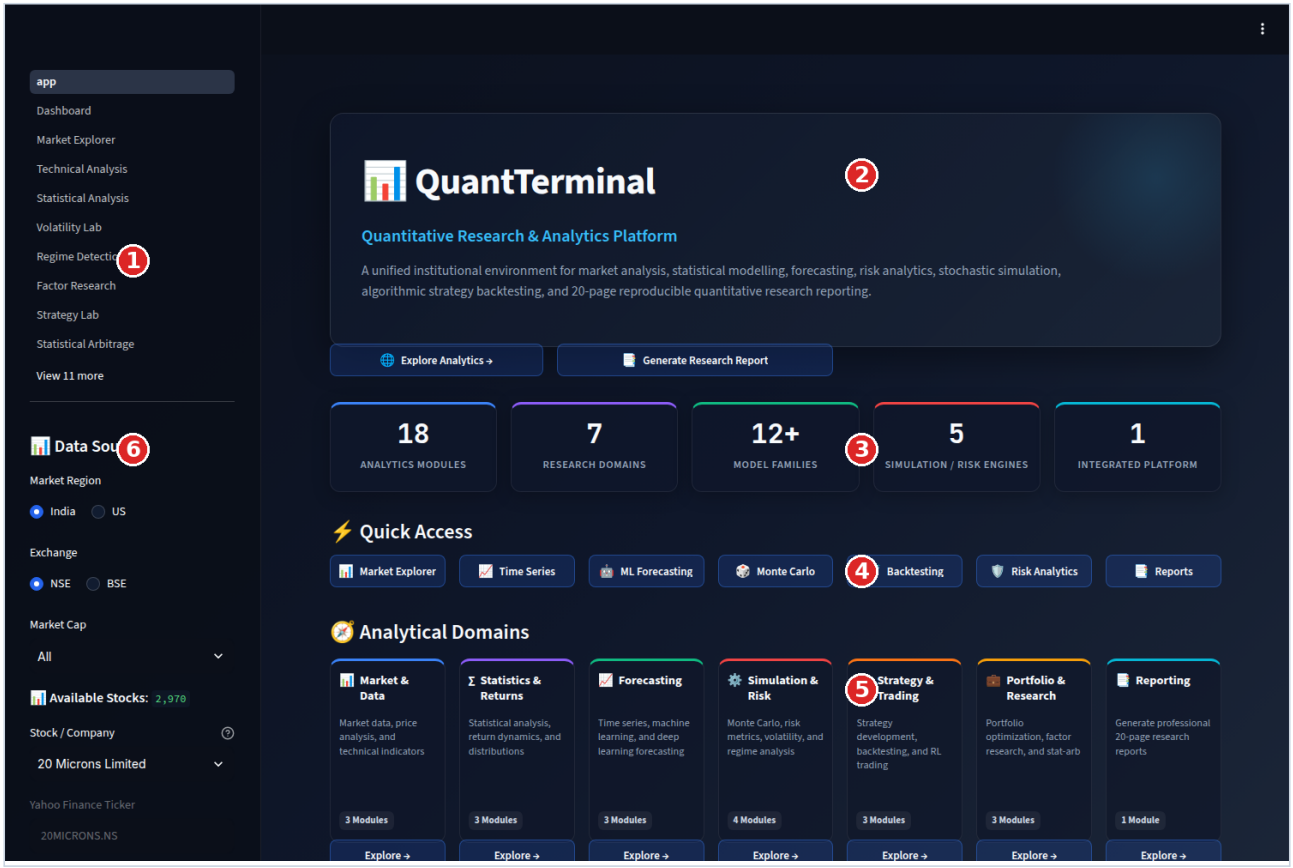


Figure 1: QuantTerminal home page (command center) from the live app

#	Area	What it does
1	Sidebar - page list	Jump to any of the 20 pages. 'View more' expands the full list.
2	Hero banner	Project name and two buttons: Explore Analytics and Generate Research Report.
3	Status cards	Quick counts: modules, domains, model families and engines.
4	Quick Access	One-click shortcuts to popular modules.
5	Analytical Domains	Seven cards. 'Explore' opens that group of modules.
6	Data Source panel	Choose region (India or US), exchange, market cap, stock, period and interval. Applies to every page.

Scroll further down on the home page to find: **Featured Research Workspaces**, **Terminal Use Cases**, the module directory with search, the capability matrix, the technology stack and the reporting spotlight. The disclaimer is shown at the bottom of every page.

7 How to Use the Terminal

Follow these six steps for any analysis.

Step	Do this	Where
1	Choose the **Market Region** (India or US) and the **Exchange** (NSE or BSE for India).	Left sidebar - Data Source
2	Optionally pick a **Market Cap** group to shorten the stock list.	Left sidebar
3	Select a **Stock / Company** from the dropdown or type a ticker such as RELIANCE.NS or AAPL.	Left sidebar
4	Choose the **Period** (for example 1Y or 5Y) and the **Interval** (1d, 1wk, 1mo).	Left sidebar
5	Open a page from the page list. The page loads data for your stock automatically.	Page list
6	Change page settings, read the charts and tables, then export (CSV, Excel or PDF) if needed.	Inside the page

Ticker format

Market	Format	Example
NSE (India)	Name followed by .NS	TCS.NS, INFY.NS
BSE (India)	Name followed by .BO	TCS.BO
US	Plain symbol	AAPL, MSFT
Indices	Start with ^	^NSEI (Nifty 50), ^GSPC (S&P 500)

Helpful habits

- Use at least 3 years of data for forecasting and 5+ years for regimes.
- Press the run or fit button on a page after changing settings.
- If a chart looks empty, try a longer period or another stock.
- Hover over charts to see exact values. Use the camera icon to save a picture.
- Look for the small 'help' question marks next to settings for quick explanations.
- Compare more than one model or setting before drawing a conclusion.

8 Recommended Workflows

Not sure where to start? Use one of these routes through the terminal.

Goal	Route	Why this order
Understand a company	Dashboard → Market Explorer → Technical Analysis → Return Analytics	See the basics first, then compare, then study price behaviour
Judge risk	Statistical Analysis → Volatility Lab → Risk Analytics → Monte Carlo	From how returns behave, to how much they swing, to how much could be lost
Test a trading idea	Technical Analysis → Strategy Lab → Backtesting → Regime Detection	Pick an idea, rank it, test it with costs, check it in each regime
Compare forecasts	Time Series → ML Forecasting → DL Forecasting → RL Trading	Go from simple to complex and compare each to buy-and-hold
Build a portfolio	Return Analytics → Factor Research → Portfolio Lab → Risk Analytics	Know each stock, know the market link, allocate, then check risk
Make a report	Run the pages you need → Report Generation	The report uses the stock and settings you selected

Worked example: a first-time research session

- Select **India**, **NSE**, and the stock **Infosys**. Set period to 5Y.
- Open **Dashboard** to see the price, P/E and returns.
- Open **Volatility Lab**. Note whether current volatility sits high or low in the cone.
- Open **Regime Detection** with 2 regimes. See how often the stock was in the stressed regime.
- Open **Backtesting**, run a simple moving-average strategy and compare it to buy-and-hold.
- Open **Report Generation** and download the PDF to keep your notes.

> Remember that each step **describes history**. None of them is a prediction or a recommendation.

9 System Architecture

QuantTerminal is built in layers. Each layer only talks to the one below it, which keeps the code tidy and testable.

Layer	Folders	Job
1. Pages (what you see)	app.py, pages/	Screens, buttons, charts
2. Shared visuals	plots/, utils/	Chart builders, theme, sidebar, formatting
3. Analysis engines	core, risk, statistics, technical, factor, regime, simulation, statarb, strategies, backtesting, optimization, portfolio, forecasting, machine_learning, ml_dl	The maths and models
4. Data	data/, services/	Loading, caching, Yahoo Finance and TradingView access, CSV snapshots
5. Reporting	reporting/, reports/	20-page PDF, CSV and Excel exports

How a request flows

1. You choose a stock	2. Page asks the data layer	3. Data layer loads or reads cache	4. Engine calculates	5. Plots draw the chart	6. You see the result

Folder map

Folder	Contents
pages/	The 20 Streamlit pages (01_Dashboard.py ... 20_Credits.py)
core/	Returns, Sharpe, Sortino, Calmar, drawdown
statistics/ and technical/	Statistical tests and indicators
strategies/ and backtesting/	11 strategies and the backtest engine
regime/ and simulation/	HMM, GMM, change points, GBM and bootstrap
ml_dl/	Training pipeline, saved models and results for ML and DL
docs/	17 markdown design documents
tests/	23 automated test files

> The full technical details live in `docs/ARCHITECTURE.md`, `docs/DATA_LAYER.md` and `docs/STREAMLIT_PAGES.md`.

10 The Data Layer

Where does the data come from?

Source	What it gives	Used by
Yahoo Finance (yfinance)	Daily and intraday prices, volume, company profile, valuation	Almost every page
TradingView screener	Sector, industry, technical ratings, screener fields	Market Explorer, Dashboard
CSV snapshots in data/snapshots	Reference list of Indian and US stocks	Stock lists in the sidebar

How data is prepared

- **Split and dividend adjusted** prices so history is comparable.
- **Calendar aligned:** holidays are dropped, not filled with invented numbers.
- **Cached:** repeated requests reuse stored data so pages open faster.
- **Status:** each data point is labelled as market data, reference snapshot or unavailable.

Supported settings

Setting	Options
Regions	India, US
Exchanges	NSE, BSE, NASDAQ, NYSE
Intervals	Daily (1d), Weekly (1wk), Monthly (1mo), Intraday (15m, 60m)
History	Up to 10 or more years of continuous daily prices
Currency	Rupee for India, dollar for US

Good to know

- Yahoo Finance is a free, unofficial source. It can be slow, limit requests or change its format.
- Intraday data only goes back a short time.
- If a stock is delisted, the page shows a clear 'Market Data Unavailable' message.

> **Rule:** QuantTerminal never creates fake prices to fill a gap. Missing data stays missing.

11 Method and Bias Controls

A bias is a hidden mistake that makes results look better than they really are. The project lists its controls in docs/BIAS_MITIGATION.md. Here are the most important ones in plain words.

Problem	Simple explanation	What QuantTerminal does
Look-ahead bias	Using tomorrow's information today	Signals are lagged by one day and features only use the past
Data leakage in ML	Test data sneaks into training	Time-ordered 80/20 split, scalers fitted on training data only
Overfitting	Model memorises the past	Out-of-sample scores, walk-forward tests, warnings on tuning
Unrealistic backtests	Ignoring costs	Brokerage and slippage are included
Spurious regression	Two trending series look related by chance	Stationarity checks before regression
Survivorship	Only looking at companies that survived	Limitations are stated clearly in the docs
Fake data	Filling gaps with invented values	No synthetic infilling

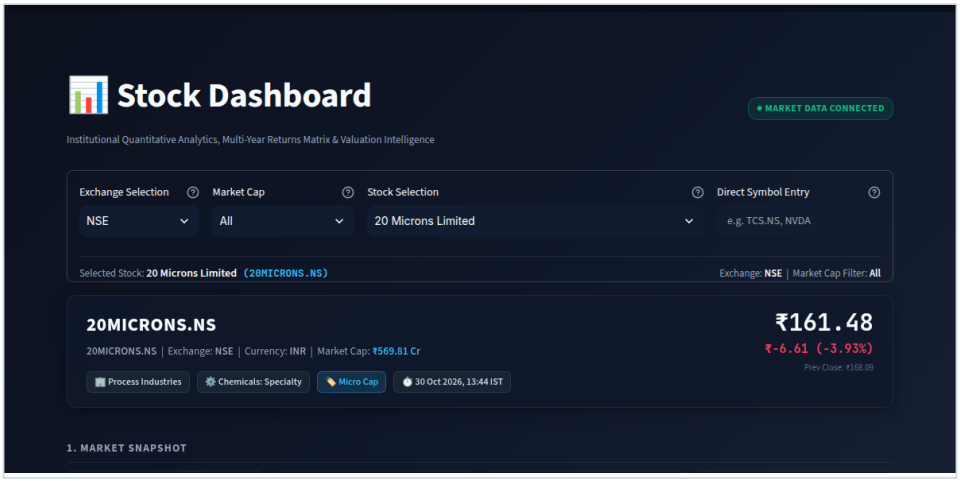
Core formulas used across the app

Measure	Plain meaning	Formula
Return	Percent change in price	$(P_t - P_{t-1}) / P_{t-1}$
Sharpe ratio	Return per unit of risk	(mean return - risk-free) / std dev x sqrt(252)
Sortino ratio	Return per unit of downside risk	(mean return - risk-free) / downside std dev
Max drawdown	Worst fall from a peak	min(equity / running peak - 1)
VaR (historical)	Loss not exceeded on most days	Percentile of past returns
CAGR	Average yearly growth	$(\text{final} / \text{initial})^{1/\text{years}} - 1$

> Even with these controls, every result is a **historical** study. Controls reduce mistakes but cannot predict the future.

Module 01 Dashboard Market & Data

Your one-page summary of a single stock. It brings price, volume, valuation numbers and returns together so you can understand a company quickly.



Dashboard page (live app)

What you can do

- See the latest price, change and trading volume
- Check valuation ratios such as P/E and market cap
- View returns over many time periods (1 month, 1 year, 5 years)
- Spot a quick momentum summary from basic indicators

How to read it

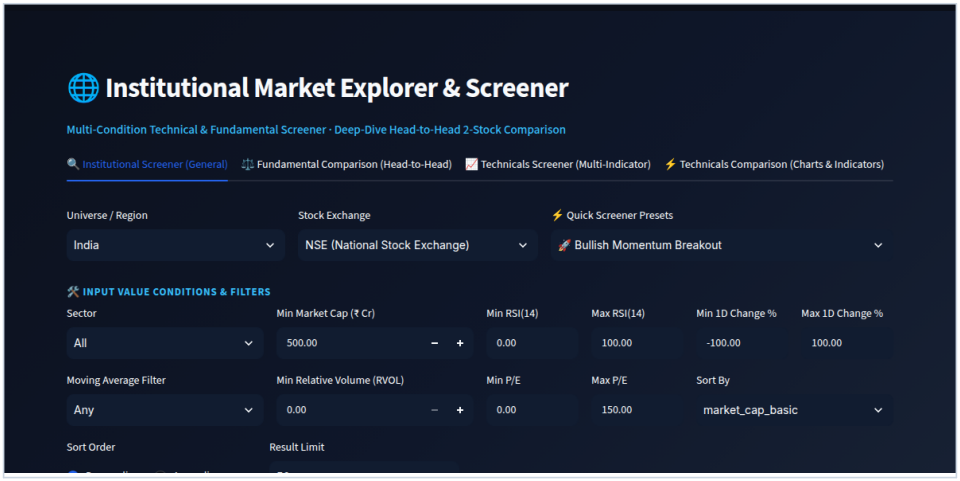
Start at the top cards, then scroll down to the returns table. Green means up, red means down.

Watch out

Numbers come from Yahoo Finance and TradingView. If a stock is delisted or the provider is slow, a 'Market Data Unavailable' box appears.

Module 02 Market Explorer Market & Data

A search and screening tool. It helps you find stocks that match your rules and compare two stocks side by side.



Market Explorer page (live app)

What you can do

- Filter stocks by market cap, sector, RSI, P/E and volume
- Use ready-made presets such as 'Bullish Momentum Breakout'
- Compare two companies head to head
- See technical ratings: Strongly Bullish, Bullish, Neutral, Bearish, Strongly Bearish

How to read it

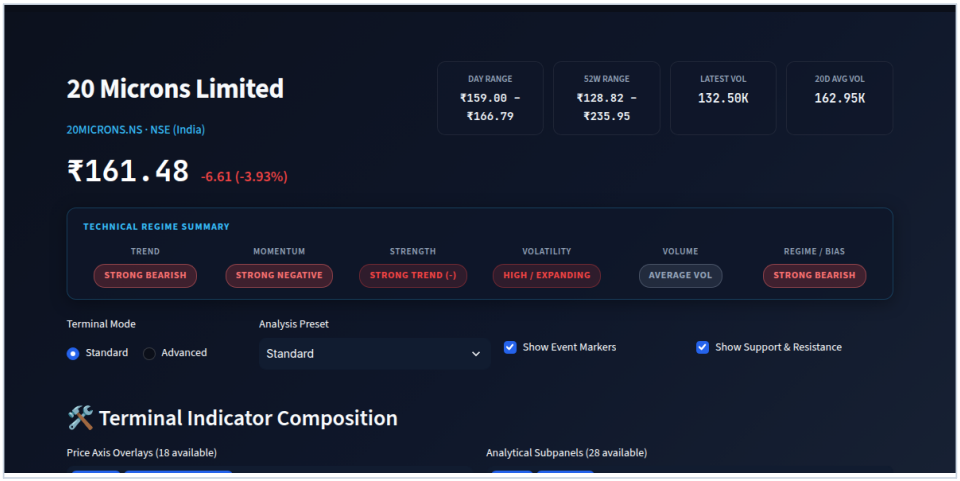
The table at the bottom lists every match. Click a column name to sort it.

Watch out

A rating is a summary of indicators, not advice. A 'Bullish' label only means the indicators lean upward.

Module 03 Technical Analysis Market & Data

A chart workstation for technical indicators. Indicators are simple formulas built from price and volume that help describe trend, speed and volatility.



Technical Analysis page (live app)

What you can do

- Draw Bollinger Bands, moving averages and SuperTrend on a candlestick chart
- Read momentum with RSI, Stochastic and MACD
- Measure volatility with ATR and Keltner channels
- Open 'playbooks' that explain how each indicator family is usually read

How to read it

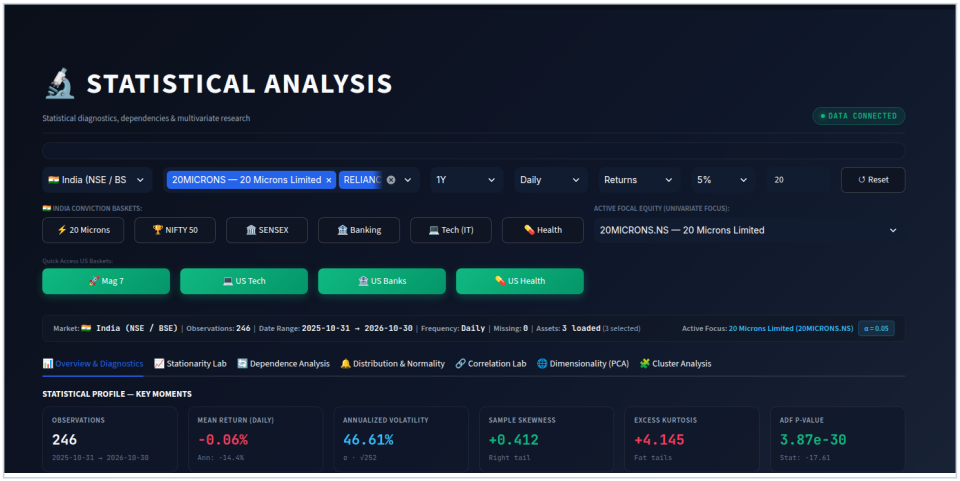
RSI above 70 is often called overbought and below 30 oversold. These are hints, not certainties.

Watch out

Indicators lag behind price and can give false signals in sideways markets. Use several together.

Module 04 Statistical Analysis Statistics & Returns

Shows how a stock's returns are spread out. It answers: are the returns 'normal', how fat are the tails, and how do stocks move together?



Statistical Analysis page (live app)

What you can do

- See skewness (lean) and kurtosis (fat tails)
- Run normality tests (Jarque-Bera, Shapiro-Wilk)
- View histograms, KDE curves and Q-Q plots
- Build correlation matrices between stocks

How to read it

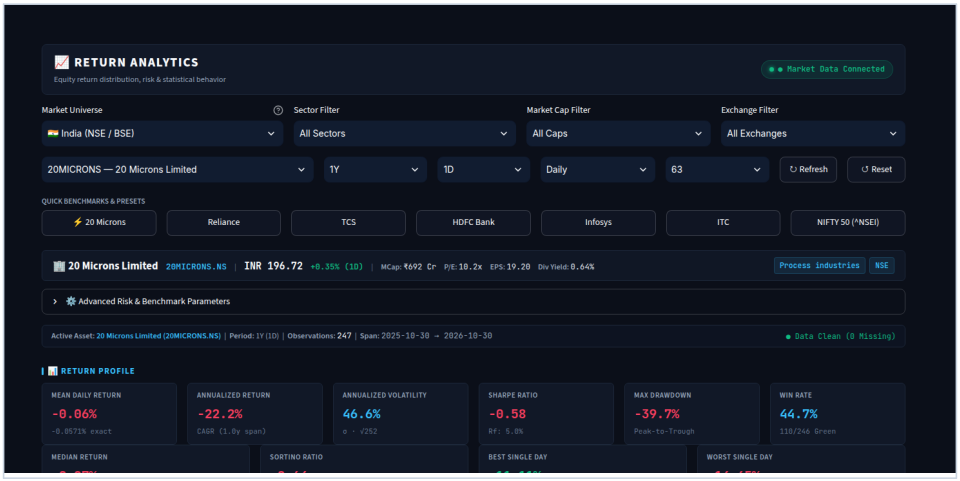
If the Q-Q plot points follow the straight line, returns look normal. Big curves at the ends mean fat tails.

Watch out

Real stock returns are rarely perfectly normal. Test results depend on how much history you pick.

Module 05 Return Analytics Statistics & Returns

Explains how money would have grown or fallen over time, with calendar views that make patterns easy to see.



Return Analytics page (live app)

What you can do

- Switch between simple and logarithmic returns
- See growth of 100 invested over time
- View monthly and yearly return heatmaps
- Check rolling 30-day and 90-day returns and volatility

How to read it

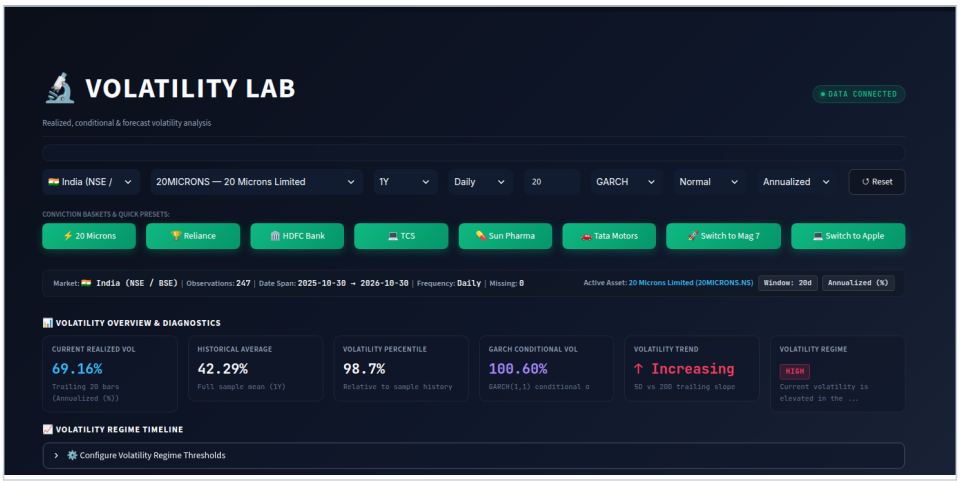
In the heatmap, deeper green is a stronger month and deeper red a weaker one.

Watch out

Past returns do not guarantee future returns. Short windows can look very different from long ones.

Module 06 Volatility Lab Statistics & Returns

Volatility means how much a price jumps around. This lab measures it in five ways and forecasts it with GARCH models.



Volatility Lab page (live app)

What you can do

- Compare 5 estimators: Close-to-Close, Parkinson, Garman-Klass, Rogers-Satchell, Yang-Zhang
- Fit GARCH(1,1) to see volatility clusters
- Use volatility cones to judge if today's volatility is high or low
- View the volatility term structure

How to read it

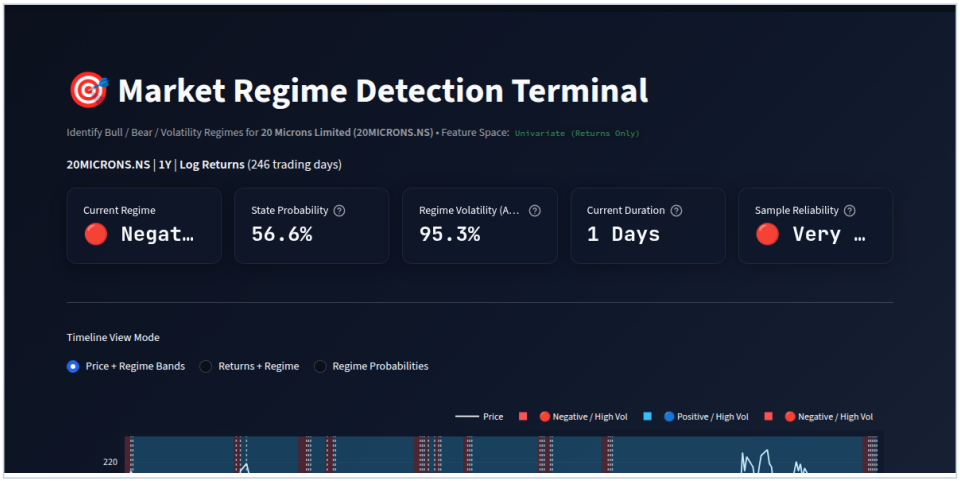
A cone shows the normal range for volatility. If today's value sits above the cone, volatility is unusually high.

Watch out

GARCH needs enough data to work well. Estimators that use high and low prices are only as good as that data.

Module 07 Regime Detection Forecasting & AI

Markets behave differently in calm and stormy times. A Hidden Markov Model (HMM) finds these hidden 'regimes' automatically.



Regime Detection page (live app)

What you can do

- Choose 2 or 3 regimes (for example calm, normal, stressed)
- See the average return and risk of each regime
- View the chance of switching from one regime to another
- Compare a regime strategy with simple buy-and-hold

How to read it

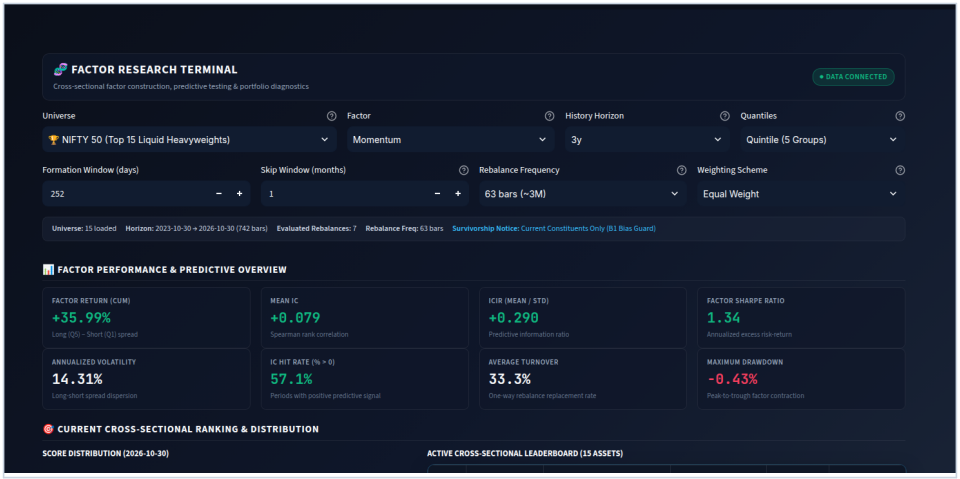
Look at the coloured price chart: each colour is one regime. The transition table tells you how long a regime tends to last.

Watch out

Regimes are labelled after the fact using the full sample, so they describe history. They are not a live prediction tool.

Module 08 Factor Research Portfolio & Research

Explains where a stock's returns come from: the overall market or something unique to the stock.



Factor Research page (live app)

What you can do

- Estimate Beta (how strongly a stock follows the market)
- Estimate Jensen's Alpha (return beyond what beta explains)
- Run style regressions inspired by Fama-French
- Watch rolling beta change over time

How to read it

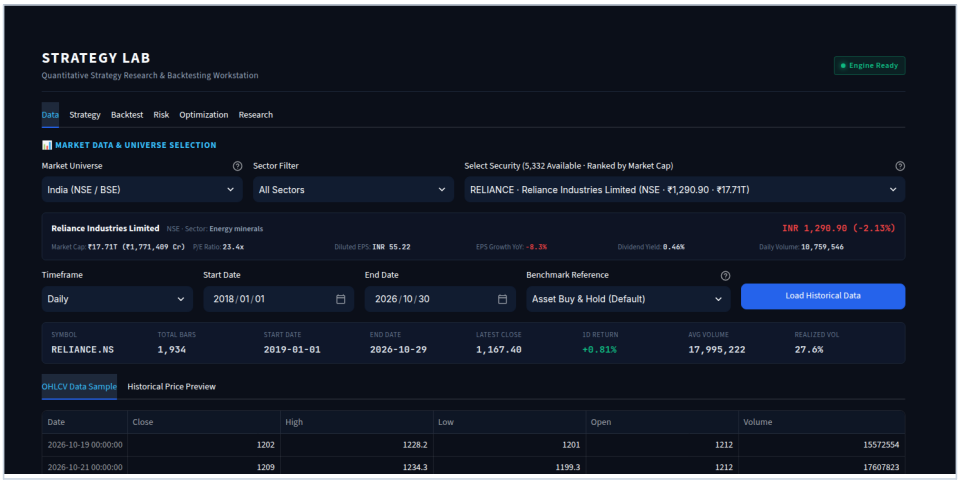
Beta of 1 moves with the market, above 1 moves more, below 1 moves less. R-squared tells you how much the market explains.

Watch out

Alpha estimates can be noisy. Check the confidence values before trusting them.

Module 09 Strategy Lab Strategy & Trading

A place to design and test trading rules. It runs 11 built-in strategies against each other in a tournament.



Strategy Lab page (live app)

What you can do

- Pick from trend, mean-reversion and breakout strategies
- Combine strategies with AND, OR or Majority Vote
- Rank all strategies on a leaderboard
- Tune settings with a 2D heatmap and walk-forward testing

How to read it

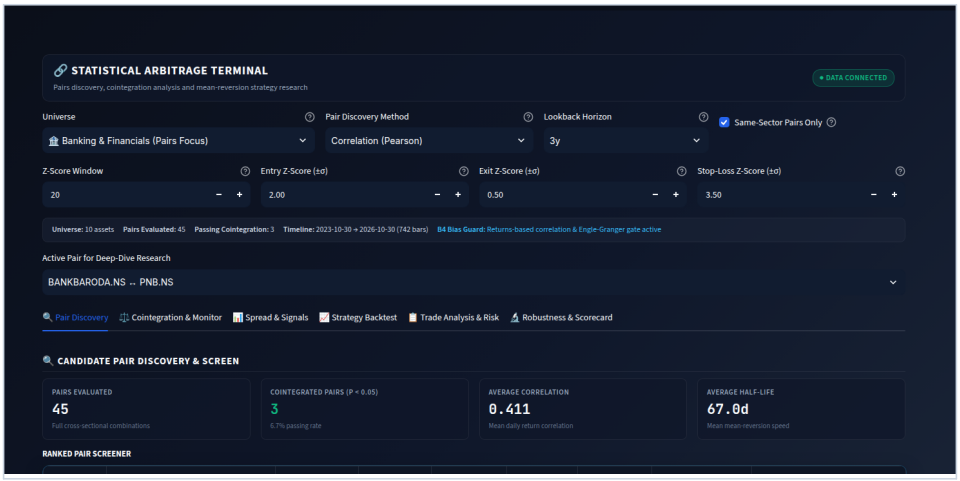
The leaderboard sorts strategies by performance and risk. Prefer strategies that stay good across many settings.

Watch out

Tuning settings too hard on past data causes overfitting. Always check walk-forward results.

Module 10 Statistical Arbitrage Strategy & Trading

Pairs trading research. It looks for two stocks that usually move together and studies when their gap becomes unusually wide.



Statistical Arbitrage page (live app)

What you can do

- Run the Engle-Granger cointegration test
- Build the spread with a hedge ratio
- Estimate the half-life (how fast the gap closes)
- See Z-score entry and exit signals and a pair backtest

How to read it

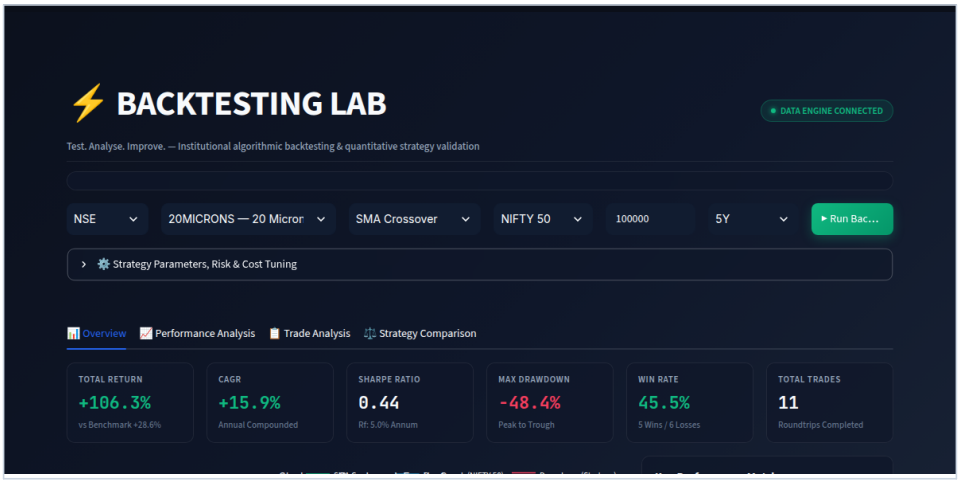
A Z-score near 0 means the pair is normal. Large positive or negative values mean the gap is stretched.

Watch out

Pairs can stop moving together. A pair that worked in the past may break in the future.

Module 11 Backtesting Strategy & Trading

A backtest replays a trading rule on past data to see how it would have done, including trading costs.



Backtesting page (live app)

What you can do

- Choose a strategy and its settings
- Add brokerage and slippage costs
- Compare against buy-and-hold or an index
- Read Sharpe, Sortino, Calmar, win rate and drawdown

How to read it

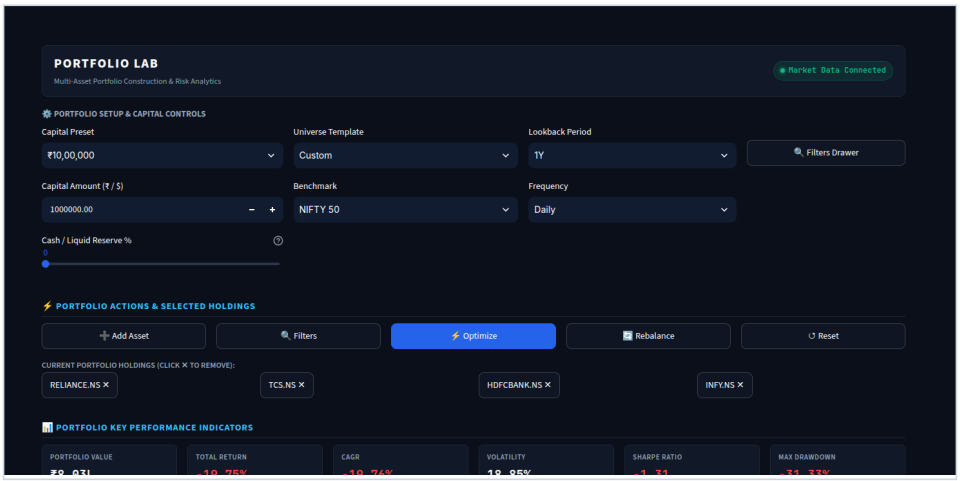
The equity curve shows growth of money. The drawdown chart shows how deep the losses went.

Watch out

A backtest uses history only. Good past results can come from luck or from over-tuning.

Module 12 Portfolio Lab Portfolio & Research

Helps you study how to split money across several stocks so risk and return are balanced.



Portfolio Lab page (live app)

What you can do

- Draw the Markowitz efficient frontier
- Find the Maximum Sharpe and Minimum Variance portfolios
- Try Risk Parity and Inverse Volatility weights
- See a rebalance table showing Bullish / Neutral / Bearish weight changes

How to read it

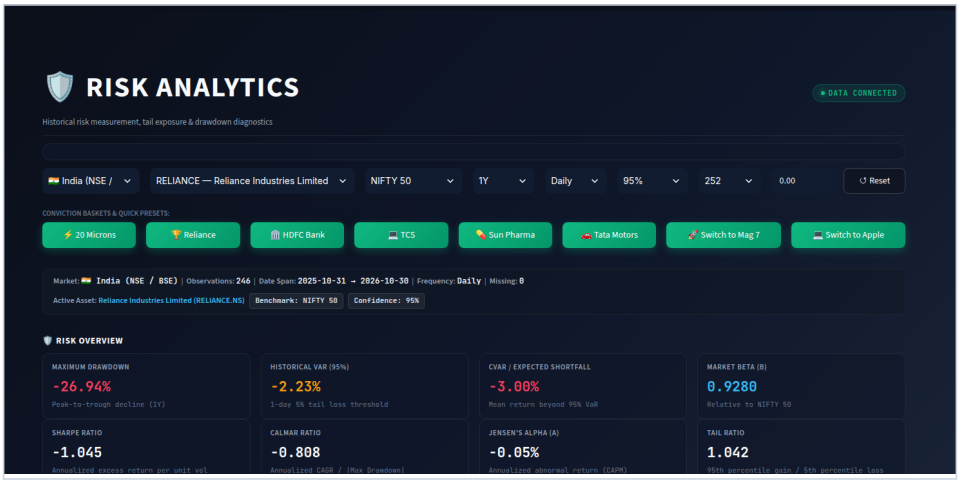
Each dot on the frontier is a portfolio. Higher means more return, further right means more risk.

Watch out

Optimisers are sensitive to input estimates. Small data changes can give very different weights.

Module 13 Risk Analytics Simulation & Risk

Measures how much you could lose. It answers 'how bad can a bad day or bad month be?'



Risk Analytics page (live app)

What you can do

- Calculate Value at Risk (VaR): Historical, Parametric, Cornish-Fisher
- Calculate CVaR (the average loss beyond VaR)
- See the underwater drawdown chart and recovery times
- Run historical stress tests

How to read it

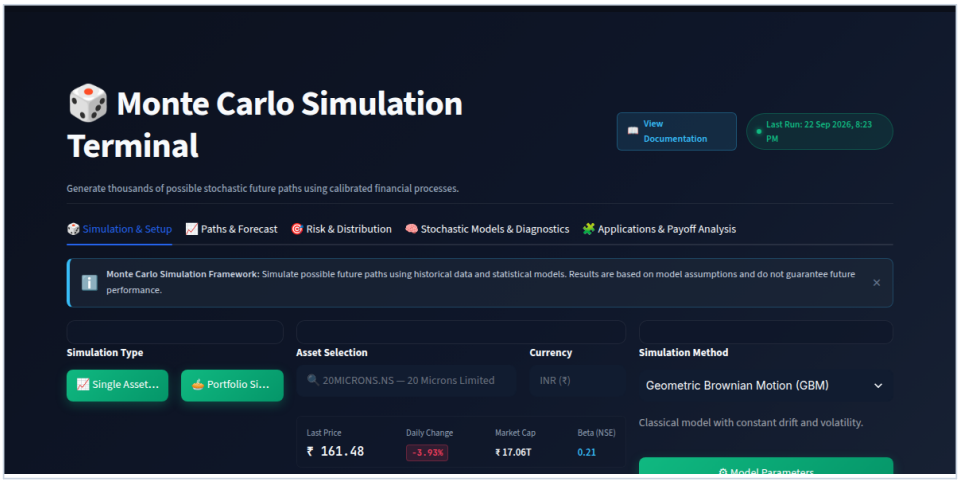
95% VaR of 2% means: on 95 out of 100 days the loss should be smaller than 2%.

Watch out

VaR says nothing about how bad the worst 5% can be. That is why CVaR is shown too.

Module 14 Monte Carlo Simulations Simulation & Risk

Creates hundreds of possible future price paths with random numbers, so you can see a range of outcomes instead of one guess.



Monte Carlo Simulations page (live app)

What you can do

- Simulate with GBM, Jump Diffusion or Bootstrap
- View a fan chart (5th to 95th percentile)
- See the distribution of final prices
- Read the probability of loss

How to read it

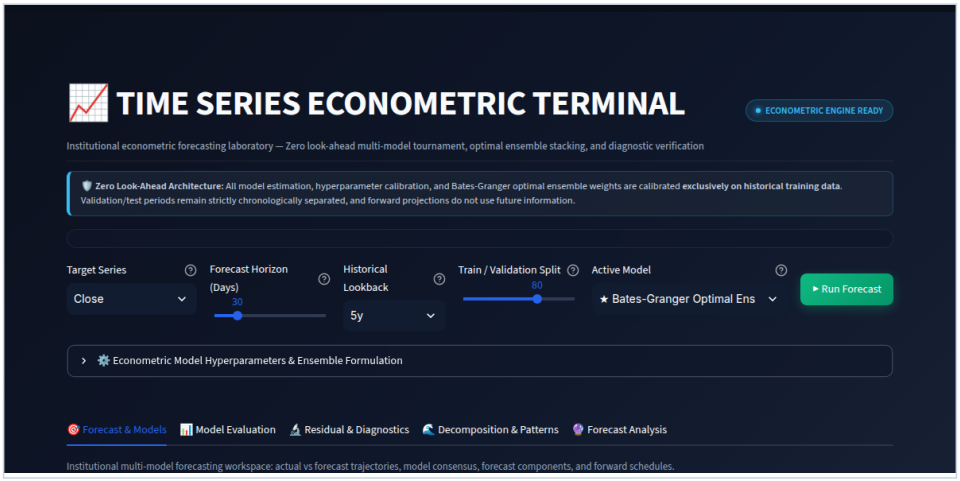
The fan shows likely ranges. The middle line is the median path, the edges are extreme cases.

Watch out

The simulation only reflects the assumptions you give it. It is not a forecast of what will happen.

Module 15 Time Series Forecasting Forecasting & AI

Classic statistics for forecasting. It fits ARIMA and SARIMA models and splits a series into trend, season and noise.



Time Series Forecasting page (live app)

What you can do

- Auto-fit ARIMA or seasonal SARIMA
- See multi-step forecasts with 95% confidence bands
- Decompose a series using STL
- Compare models with AIC, BIC and HQIC

How to read it

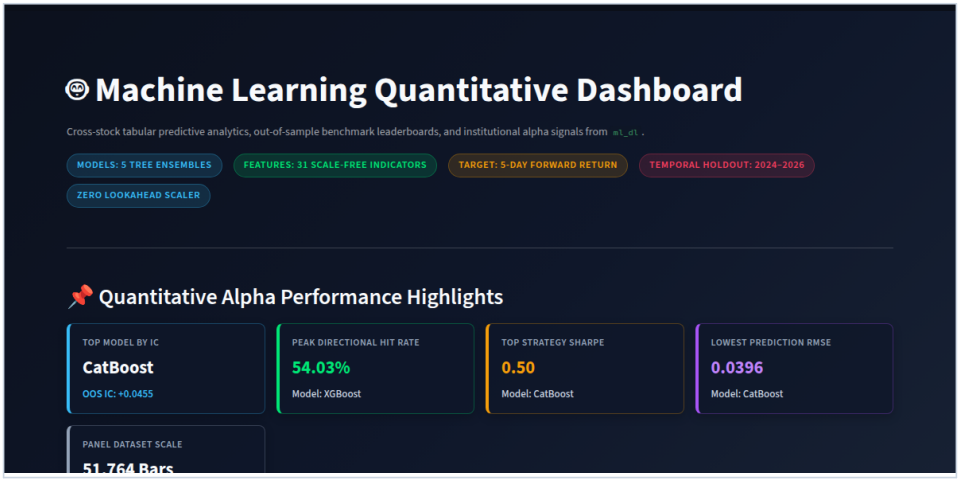
The shaded band around the forecast shows uncertainty. A wider band means less confidence.

Watch out

Forecast bands widen quickly. Treat long forecasts with great care.

Module 16 ML Forecasting Forecasting & AI

Uses machine learning models (Linear, Ridge, Lasso, Random Forest, XGBoost and more) to predict returns from engineered features.



ML Forecasting page (live app)

What you can do

- Compare many models on one scoreboard
- See which features matter most
- Test on a time-ordered 80/20 split (no peeking at the future)
- Check IC, hit rate, RMSE and MAE
- Read the model's Bullish, Bearish or Neutral outlook

How to read it

Compare models using out-of-sample numbers, not training numbers. A high hit rate near 50% is no better than a coin flip.

Watch out

Markets are noisy. Models often fail to beat simple benchmarks, so treat results as research.

Module 17 DL Forecasting Forecasting & AI

Deep learning for sequences. LSTM and GRU networks read many days of history at once to estimate what comes next.



DL Forecasting page (live app)

What you can do

- Choose LSTM or GRU
- Watch loss curves to see training progress
- See out-of-sample predictions
- Compare against buy-and-hold

How to read it

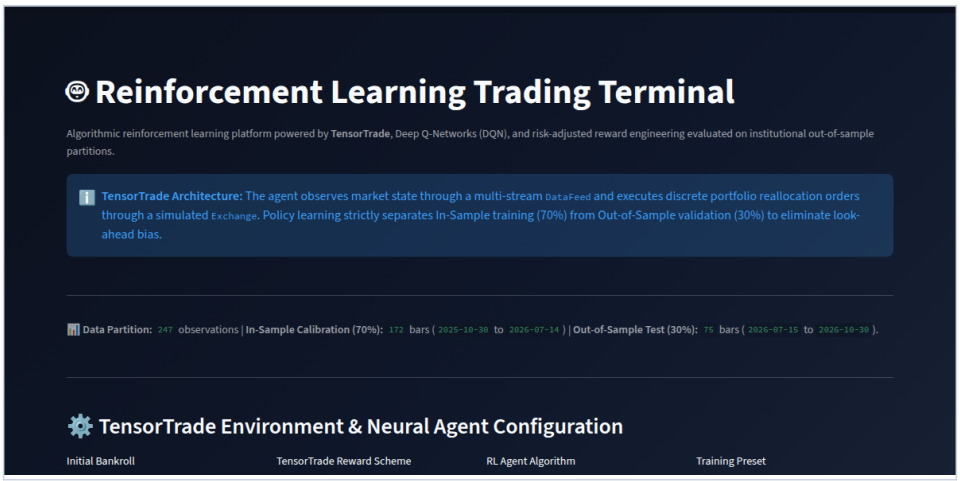
If training loss keeps falling but test loss rises, the model is memorising (overfitting).

Watch out

Deep models need a lot of data and are hard to explain. Results can change between runs.

Module 18 RL Trading Forecasting & AI

A learning agent that tries actions in a simulated market and gets rewards. Over time it learns a policy.



RL Trading page (live app)

What you can do

- Train agents such as DQN, PPO and SAC
- Set reward style (profit, Sharpe, Sortino)
- See the agent's Bullish, Bearish or Neutral stance over time
- Compare with buy-and-hold and export the order log

How to read it

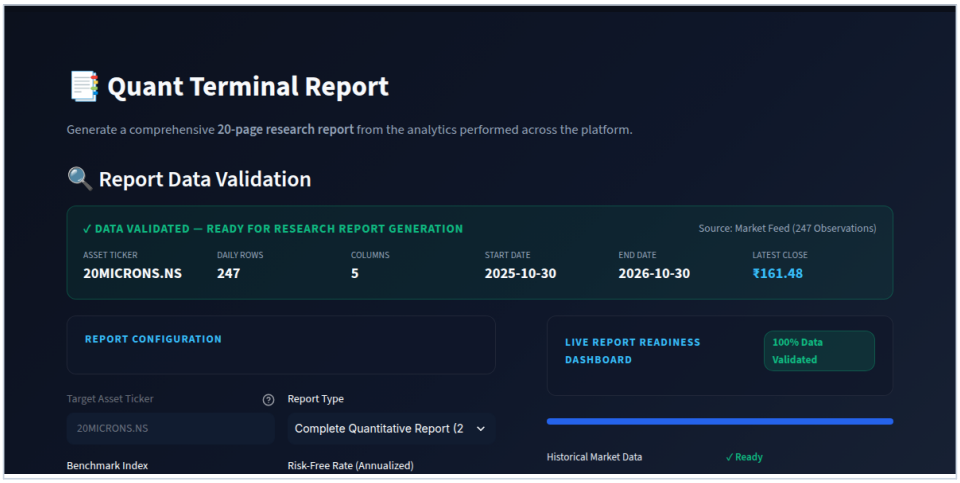
The action chart shows when the agent leaned Bullish or Bearish. Check the out-of-sample section, not the training section.

Watch out

Training is slow and unstable. A simulated market is not the real market.

Module 19 Report Generation Reporting

Builds a 20-page research report as a PDF for the selected stock, with charts, tables and notes.



Report Generation page (live app)

What you can do

- Pick the stock and sections
- Generate with one click and download
- Get charts for returns, risk, forecasts and simulations
- Use it for coursework or notes

How to read it

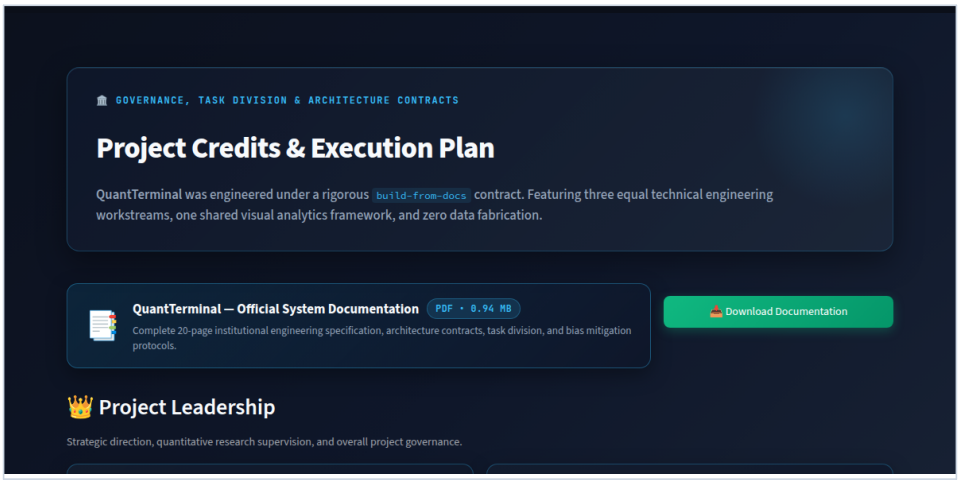
Read the summary page first, then go to the sections you care about.

Watch out

Reports are generated from the same data you see in the app. Check the data date at the top.

Module 20 Credits Reporting

The project page: team, roles, build timeline and downloads of the system documentation.



Credits page (live app)

What you can do

- See who built what
- View the 30-day project plan
- Download the system documentation PDF
- Read the educational-use disclaimer

How to read it

Use it to understand the project structure and where each module came from.

Watch out

This page is informational. It has no analysis tools.

Module Quick-Reference Table

Use this page to pick the right module fast.

Module	Domain	One-line summary
Dashboard	Market & Data	Your one-page summary of a single stock.
Market Explorer	Market & Data	A search and screening tool.
Technical Analysis	Market & Data	A chart workstation for technical indicators.
Statistical Analysis	Statistics & Returns	Shows how a stock's returns are spread out.
Return Analytics	Statistics & Returns	Explains how money would have grown or fallen over time, with calendar views that make patterns easy to see.
Volatility Lab	Statistics & Returns	Volatility means how much a price jumps around.
Regime Detection	Forecasting & AI	Markets behave differently in calm and stormy times.
Factor Research	Portfolio & Research	Explains where a stock's returns come from: the overall market or something unique to the stock.
Strategy Lab	Strategy & Trading	A place to design and test trading rules.
Statistical Arbitrage	Strategy & Trading	Pairs trading research.
Backtesting	Strategy & Trading	A backtest replays a trading rule on past data to see how it would have done, including trading costs.
Portfolio Lab	Portfolio & Research	Helps you study how to split money across several stocks so risk and return are balanced.
Risk Analytics	Simulation & Risk	Measures how much you could lose.
Monte Carlo Simulations	Simulation & Risk	Creates hundreds of possible future price paths with random numbers, so you can see a range of outcomes instead of one guess.
Time Series Forecasting	Forecasting & AI	Classic statistics for forecasting.
ML Forecasting	Forecasting & AI	Uses machine learning models (Linear, Ridge, Lasso, Random Forest, XGBoost and more) to predict returns from engineered features.
DL Forecasting	Forecasting & AI	Deep learning for sequences.
RL Trading	Forecasting & AI	A learning agent that tries actions in a simulated market and gets rewards.
Report Generation	Reporting	Builds a 20-page research report as a PDF for the selected stock, with charts, tables and notes.
Credits	Reporting	The project page: team, roles, build timeline and downloads of the system documentation.

12 Glossary of Terms

Simple meanings of words used in this document.

Term	Meaning
Alpha	Return beyond what the market explains
ARIMA	A statistics model that forecasts from past values and past errors
ATR	Average True Range: a measure of typical daily price movement
Backtest	Replaying a rule on past data
Beta	How strongly a stock moves with the market
Bollinger Bands	Lines drawn around a moving average based on volatility
CAGR	Compound yearly growth rate
CVaR	Average loss on the worst days beyond VaR
Cointegration	Two prices that drift together over time
Drawdown	Fall from a peak to a later low
Fat tails	Extreme moves happen more often than a bell curve predicts
GARCH	Model where volatility depends on recent shocks
HMM	Hidden Markov Model: finds hidden states such as calm or stormy markets
IC	Information Coefficient: correlation of predictions with real results

Term	Meaning
KDE	Smooth curve estimate of a histogram
LSTM / GRU	Neural networks that read sequences
MACD	Indicator comparing a fast and a slow moving average
Monte Carlo	Many random simulations to see a range of outcomes
Out-of-sample	Data the model has never seen
Overfitting	Learning noise instead of the real pattern
RSI	Relative Strength Index from 0 to 100
Sharpe ratio	Return per unit of risk
Slippage	Difference between expected and actual trade price
Sortino ratio	Like Sharpe but only counts downside risk
VaR	Value at Risk: a loss level not usually exceeded
Volatility	How much a price moves up and down
Z-score	How many standard deviations away from average

13 Capability Matrix and Technology Stack

Module	Analysis	Model	Simul.	Backtest	Risk	Charts	Export
Dashboard	Yes	-	-	-	Yes	Yes	Yes
Market Explorer	Yes	-	-	-	Yes	Yes	Yes
Technical Analysis	Yes	-	-	-	-	Yes	Yes
Return Analytics	Yes	-	-	-	Yes	Yes	Yes
Statistical Analysis	Yes	Yes	-	-	-	Yes	Yes
Volatility Lab	Yes	Yes	-	-	Yes	Yes	Yes
Regime Detection	Yes	Yes	-	Yes	Yes	Yes	Yes
Factor Research	Yes	Yes	-	-	Yes	Yes	Yes
Strategy Lab	Yes	Yes	-	Yes	Yes	Yes	Yes
Statistical Arbitrage	Yes	Yes	-	Yes	-	Yes	Yes
Backtesting	Yes	-	-	Yes	Yes	Yes	Yes
Portfolio Lab	Yes	Yes	Yes	-	Yes	Yes	Yes
Risk Analytics	Yes	-	Yes	-	Yes	Yes	Yes
Monte Carlo	Yes	Yes	Yes	-	Yes	Yes	Yes
Time Series	Yes	Yes	-	-	-	Yes	Yes
ML Forecasting	Yes	Yes	-	Yes	-	Yes	Yes
DL Forecasting	Yes	Yes	-	Yes	-	Yes	Yes
RL Trading	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Report Generation	Yes	-	-	-	-	Yes	Yes

Layer	Technology
Language	Python 3.10 - 3.12
Web framework	Streamlit
Data	pandas, numpy, yfinance, pyarrow
Charts	Plotly (interactive), Matplotlib (report charts)
Maths and models	scipy, statsmodels, arch, scikit-learn, XGBoost, LightGBM, CatBoost, hmmlearn, ruptures
Deep learning and RL	TensorFlow, Keras, Gymnasium, TensorTrade
Documents	ReportLab, openpyxl, pypdf, PyMuPDF
Hosting	Streamlit Community Cloud

14 Testing, Limitations and Roadmap

Testing

The project has **23 automated test files** (run with `pytest`). They check the core maths and exports:

Area	Test files
Returns, metrics, drawdown	test_returns, test_metrics, test_drawdown
Risk and volatility	test_risk_metrics, test_rolling, test_garch, test_estimators
Statistics	test_summary, test_distributions, test_correlation, test_pca, test_clustering, test_stationarity, test_diagnostics, test_timeseries
Factors and pairs	test_factors, test_scores, test_pairs, test_cointegration, test_spread
Exports	test_csv, test_excel, test_pdf, test_reporting_pipeline

Honest limitations

- The pages themselves (the screens) are not covered by automatic tests.
- Strategies, optimisation, regime and simulation code have fewer direct tests.
- Data comes from free providers that may be slow or change without notice.
- Large page files make the code harder to maintain.
- Machine learning and deep learning results can change between runs and rarely beat simple baselines by much.
- Regimes are labelled with the whole sample, so they describe history.

Roadmap ideas

- Add tests for every page and engine.
- Split the biggest pages into smaller parts.
- Add more data sources and a lock file for libraries.
- Add more markets, factor models and explanation tools for ML.
- Add user accounts to save settings and reports.

15 FAQ, Troubleshooting and Appendix

Question	Answer
Does the app tell me what to buy?	No. It only shows Bullish, Bearish or Neutral readings of data. It is for education.
The page shows 'Market Data Unavailable'.	Check the ticker spelling and suffix (.NS or .BO). Try again later in case the provider is limiting requests.
Charts are empty.	Use a longer period or a more traded stock. Very new stocks have little history.
The app is slow.	ML, DL and RL pages train models. Use a shorter period or fewer epochs.
The live app shows a loading screen.	It may have been asleep. Wait up to a minute.
Which markets work?	NSE, BSE, NYSE and NASDAQ stocks and common indices.
Can I save my work?	Export tables as CSV or Excel, and use Report Generation for a PDF.
Why do results change between runs?	Neural networks and simulations use random numbers.

Appendix: documents in the docs folder

File	Topic
ARCHITECTURE.md	Folder tree and function contracts
DATA_LAYER.md	Loader, cache and snapshots
BIAS_MITIGATION.md	Bias controls
STATISTICAL_MODELS.md, RISK_ANALYTICS.md	Statistics and risk formulas
STRATEGIES_BACKTESTING.md	Strategies and backtest metrics
FORECASTING_ML.md, REGIME_SIMULATION.md	Forecasting, regimes, simulation
PORTFOLIO_OPTIMIZATION.md, FACTOR_RESEARCH.md	Allocation and factor models
REPORTS_EXPORT.md, STREAMLIT_PAGES.md	Exports and page descriptions

Final reminder: This dashboard is for educational purposes only. We are not liable for any investment decisions made by users.